

# Aditya Kumar Gupta

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## SUMMARY

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Final-year Information Science student skilled in Python, SQL, AI/ML, and data analytics with experience building AI-powered validation and prediction systems. Proficient in developing backend APIs, data pipelines, and analytics solutions using FastAPI, Scikit-learn, Pandas, and LLM workflows. Experienced in extracting actionable insights, automating workflows, and improving system reliability. Passionate about building AI-powered, data-driven solutions that improve decision-making and reliability.

## WORK EXPERIENCE

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**Android Developer Intern — MindMatrix.io** Feb 2026 – May 2026

- Built a hyperlocal worker assistance Android app using Kotlin, Jetpack Compose, MVVM, and Firebase.
- Developed realtime worker discovery, booking, ratings, and profile management features.
- Implemented worker dashboards, availability tracking, and Firebase-based realtime synchronization.

## PROJECTS

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**AI Self-Healing Validation System — Python, FastAPI, LLM, LangGraph** [GitHub](#) — [Demo](#)

- Built an AI-driven system to detect and correct incorrect outputs using LLM-based validation pipelines.
- Designed a multi-step workflow (validation → correction → re-evaluation) to improve response reliability.
- Integrated LLM APIs with backend services using FastAPI for automated issue detection and resolution.
- Reduced manual intervention by ~30% by implementing iterative self-healing mechanisms.

**Smart Crop Yield Prediction System — Python, FastAPI, Scikit-Learn** [GitHub](#) — [Demo](#)

- Built a FastAPI-based REST API to process multi-variate agricultural inputs for crop yield prediction.
- Developed a Random Forest model trained on 43,000+ data points across crops and regions.
- Designed preprocessing pipelines using ColumnTransformer with scaling, encoding, and data handling.
- Built an interactive dashboard for visualizing predictions and data-driven insights.

## EDUCATION

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|-------------|---------------------------------------------------------------|------------------|
| 2022 - 2026 | B.E at Sir M. Visvesvaraya Institute of Technology, Bengaluru | (GPA: 8.35/10.0) |
| 2019 - 2021 | Intermediate(CBSE) at Holy Mission High School, Samastipur    | (Percent: 70.8)  |
| 2018 - 2019 | Matriculation(CBSE) at St. Xaviers High School, Samastipur    | (Percent: 77)    |

## TECHNICAL SKILLS

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|                                     |                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Languages:                          | Python, SQL (MySQL), JavaScript, HTML, CSS                                                                                     |
| Backend & APIs:                     | FastAPI, RESTful APIs, JWT Authentication, Pydantic                                                                            |
| Data Analytics & ML:                | Scikit-learn (Random Forest), Pandas, NumPy, Feature Engineering, Data Preprocessing, Model Evaluation (RMSE, R <sup>2</sup> ) |
| AI & LLMs:                          | LangGraph, LLM APIs, AI Coding Tools                                                                                           |
| Data Visualization & Core Concepts: | Power BI, Chart.js, Git, Docker, OOP, DBMS, OS, CN, DSA                                                                        |

## CERTIFICATIONS / VIRTUAL EXPERIENCE PROGRAMS

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**JPMorgan Chase & Co. – Quantitative Research Virtual Experience (Forage)** Oct 2025

- Analyzed loan data to estimate default probabilities using quantitative research methods.